

Final Project: Object Detection and Counting

SHINE-MING WU SCHOOL OF INTELLIGENT ENGINEERING
2025-2026 Spring Deep Learning Final Project

1 Overview

In this project, you will form a group of **two to three** students to collaboratively build an image understanding system for animal recognition and counting. Although the core visual detector is based on YOLO, the task is designed as a structured multimodal information extraction problem: the system must convert input images into a clean, machine-readable natural language label dictionary.

The final project consists of three parts.

In the first part, teams are encouraged to build or collect training data to the greatest extent possible to improve their model's performance. Teams may use public datasets, self-collected images, data augmentation, human annotation, or a combination of these methods. The training data does not need to be submitted as a formal evaluation dataset, but the report should clearly describe the data collection, cleaning, annotation, and how it was used in training or fine-tuning.

The second part involves building an animal detection and counting system. This system takes a single image as input and outputs a Python-style dictionary. The keys should be the animal categories detected in the image, and the values should be the count of each animal category. The system should only include animal categories that appear in the image. Therefore, every value in the output dictionary must be an integer greater than or equal to 1.

The third part is evaluation. The instructor will first provide a small validation set of approximately **10 images** so that each group can test whether their data processing workflow and output format are functioning correctly. For the final grading, the instructor will use an independent synthetic evaluation set generated from predefined animal categories. Each group must process the complete evaluation set using their system and submit a JSON prediction file.

Finally, you must submit a report written in Word. The report should include the following sections: Introduction, Methodology (methods for constructing or collecting training data, technical implementation details of the YOLO-based animal counting system), and Analysis (challenges encountered, solutions developed, and error analysis, etc.).

2 YOLO System Implementation

We recommend using **YOLO** to build the animal detection and counting system, though you are free to explore other object detection methods if the final output satisfies the required dictionary format.

2.1 Details of the Animal Detection and Counting Task

The goal of this task is to recognize animals in an image and count how many instances of each animal category are present. The input is one image. The output must be a dictionary:

```
1 {"cat": 2, "duck": 1, "deer": 1}
```

This example means that the system detects two cats, one duck, and one deer in the image. The output must follow the following rules:

- **Animal-only constraint:** the system should only output animal categories. Non-animal objects such as person, car, chair, bowl, tree, or cage must not appear in the dictionary.
- **Positive-count constraint:** each output count must be an integer greater than or equal to 1.
- **Observed-category constraint:** only detected animal categories should be written into the dictionary. Do not include categories with count 0.
- **Single-image input:** each prediction is made for one image independently.
- **Multiple-species support:** one image may contain several different animal species, and the system must count each detected species separately.

2.2 Details of YOLO

YOLO, short for “You Only Look Once”, is a real-time object detection framework. Instead of first generating region proposals and then classifying each region, YOLO predicts bounding boxes and class probabilities directly from the full image in a single forward pass. This makes YOLO efficient and suitable for time-limited evaluation.

For this project, a typical YOLO-based pipeline includes the following steps:

- **Image loading:** read the input image and convert it into the format required by the detector.
- **Object detection:** run a pretrained or fine-tuned YOLO model to obtain bounding boxes, confidence scores, and predicted class labels.
- **Animal filtering:** remove all detections whose class labels are not animal categories.
- **Confidence filtering:** discard predictions whose confidence scores are below a selected threshold.
- **Non-maximum suppression:** remove duplicated bounding boxes that refer to the same animal instance.
- **Counting:** group the remaining detections by animal category and count the number of instances in each group.
- **Dictionary formatting:** output a clean dictionary containing only animal categories with positive counts.

2.3 Required System Features

Each group should implement the following features:

- **Animal category whitelist:** define a set of supported animal labels. If using a pre-trained YOLO model, groups may need to map its labels to the evaluation categories. If using a custom dataset, the model should be trained or fine-tuned to recognize the evaluation categories listed below.
- **Robust output parser:** ensure that the final result is a valid dictionary. The system should not output natural-language explanations during evaluation.
- **Threshold control:** tune the confidence threshold to balance missing animals and false detections.
- **Duplicate handling:** use non-maximum suppression or model-provided post-processing to avoid counting the same animal more than once.
- **Batch evaluation script:** provide a script that reads all evaluation images, runs the system, and writes predictions to a JSON file.

2.4 Evaluation Animal Categories

The evaluation set will contain animals from the following category list. The final answer should use these English category names as dictionary keys.

cat	dog	horse	cow
sheep	goat	pig	rabbit
chicken	duck	goose	deer
monkey	fox	wolf	bear
tiger	lion	zebra	giraffe

If an image contains animals outside this list, groups should not output unsupported category names unless explicitly instructed during evaluation.

3 Evaluation

The evaluation process consists of four parts:

- A small validation set containing approximately **10 images**, released prior to the final evaluation and used solely for system debugging and format checking.
- The final evaluation uses an independent synthetic evaluation set generated by the instructor.
- Group presentations

- Review of reports and code to determine whether the training data construction and model design are reasonable.

The validation set is released to students in advance and contains approximately **10 images**. Its purpose is to help teams check image loading, animal filtering, dictionary formatting, and JSON writing. The validation set is not the final grading set.

Although teams are not required to submit a training dataset, they are encouraged to construct diverse training data. It is recommended to include the following:

- Images containing only a single animal;
- Images containing multiple animals of the same category;
- Images containing animals from multiple categories;
- Images with partial occlusions;
- Images containing small animals or animals at a distance;
- Images containing background objects that should not be counted.

For the final evaluation, each group should process the full evaluation set and submit one JSON file. The prediction file should use the following format:

```

1 {
2   "eval_000001.png": {"cat": 2, "duck": 1, "deer": 1},
3   "eval_000002.png": {"cat": 2, "bear": 2, "deer": 1},
4   "eval_000003.png": {"dog": 2, "duck": 1, "deer": 2}
5 }
```

The validation and evaluation sets are different from groups' self-constructed training data. Each group should take the evaluation images as input and record the system's responses in a **JSON file**. Natural-language explanations should not be included in the JSON file.

Scoring Rule. Each test sample is worth **100 points**. Suppose the ground truth contains N animal categories. The 100 points are evenly divided among these N categories, so each category is worth $\frac{100}{N}$ points.

For each ground-truth animal category:

- If the category is correctly recognized, the group receives half of that category's points: $\frac{50}{N}$.
- If the predicted count for that category is also correct, the group receives the remaining half: $\frac{50}{N}$.

If the submitted prediction contains an animal category that does not appear in the ground truth, **5 points** will be deducted for each extra wrong category. The minimum score for each sample is 0, so deductions stop once the sample score reaches 0.

For example, if the ground truth is {"cat": 2, "duck": 1, "deer": 1}, then $N = 3$, and each category is worth $\frac{100}{3}$ points. Correctly detecting duck gives $\frac{50}{3}$ points, and correctly counting one duck gives another $\frac{50}{3}$ points. If the model also incorrectly predicts {"wolf": 1}, 5 points are deducted from the sample score.

Evaluation Metrics. The system will be evaluated by both dictionary validity and counting accuracy:

- **Format validity:** whether the output is a valid dictionary and contains only animal categories.
- **Category precision:** whether predicted animal categories actually appear in the image.
- **Category recall:** whether all animal categories in the reference answer are detected.
- **Counting accuracy:** whether the predicted count for each animal category matches the reference count.
- **Overall response quality:** whether the system can process all evaluation images and return a complete JSON prediction file.

4 Grading Breakdown

The scoring criteria for the final project evaluation are as follows:

- **Report Quality (20%):** The quality of the written report accounts for 20% of the total project score. This section evaluates the clarity, completeness, and professionalism of the submitted documentation.
- **Workload (20%):** The workload involved in data construction and model development accounts for 20% of the total score.
- **Response Accuracy (30%):** The validity and accuracy of the system's JSON predictions account for 30% of the final score.
- **Code (10%):** The organization, readability, reproducibility, and efficiency of the code account for 10% of the total score.
- **Presentation (10%):** The overall performance of the in-person group presentation (including speech performance and PowerPoint quality) accounts for 10% of the total score.
- **Novelty (10%):** The innovativeness of the system implementation and dataset construction accounts for 10% of the final score.

5 Submission

Each group is required to complete all tasks described in the previous sections, including the **Presentation PPT**, the **source codes** and the **project report**. The report must include the name and student ID of every group member, as well as a clear description of each member’s contribution to the project.

The group leader must submit all required materials through **Blackboard**. The submission should be packaged into a single zip archive named:

`StudentName-1_StudentName-2_project.zip`

The zip archive must contain the following files:

- Project report:

`StudentName-1_StudentName-2_project_report.pdf`

- Source code package:

`StudentName-1_StudentName-2_project_code.zip`

- Presentation PPT:

`StudentName-1_StudentName-2_project_presentation.ppt`

Task	Deadline	Time
Release of evaluation image set (provided by TA)	Jun 14	14:00
Submission of evaluation response JSON file	Jun 14	14:30
Announcement of final accuracy results (provided by TA)	Jun 15	23:59
Project presentation	Week 16	19:00
Final submission of Presentation PPT, report and source code	Jun 28	19:00

Table 1: Schedule for Project Submission

Table 1 summarizes the important deadlines and milestones for the project. All groups are expected to complete model training, prepare the required documents, and submit the evaluation responses according to the specified schedule.

The evaluation image set will be released by the teaching assistants at 14:00 on Jun 14. After receiving the evaluation data, groups must generate predictions and submit the corresponding JSON response file before 14:30 on the same day. The final accuracy results will be announced by the teaching assistants before Jun 15, 23:59.

In addition, each group is required to give a final presentation during the Deep Learning lesson in the 16th week to demonstrate their methodology, experimental results, and project outcomes. The final version of the report together with the complete source code must be submitted no later than Jun 28, 19:00.

Please ensure that:

1. All files are correctly named according to the required format;
2. The submitted code is executable and properly organized;
3. The report clearly explains the methodology, experiments, and results;
4. All submissions are uploaded before the deadlines.

Late submissions may receive penalties according to the course policy. Therefore, groups are strongly encouraged to prepare and verify all materials in advance to avoid submission issues.